

Name _____ Date _____ Period _____

Composition to Verify Inverses

Algebra II Honors | Unit 1 | Day 13 | TEK A2.2.D

Objective: I can compose two functions, use $f(g(x)) = x$ and $g(f(x)) = x$ to prove they are inverses, and state the domain restriction a non-linear pair requires.

Vocabulary

Composition — one function inside another, $f(g(x))$ or $(f \circ g)(x)$. Evaluate from the _____.

The inverse test — f and g are inverses only when _____ AND _____. One direction is not enough.

Order matters — in general $f(g(x))$ _____ the same as $g(f(x))$.

Part 1 — Warm Up and Prove

Let $f(x) = 3x - 5$ and $g(x) = (x + 5)/3$.

$f(4) =$ _____ $g(7) =$ _____ $f(g(4)) =$ _____ $g(f(4)) =$ _____

Now prove it in general, both directions.

$f(g(x)) = 3 \cdot (x+5)/3 - 5 =$ _____ $g(f(x)) = ((3x-5)+5)/3 =$ _____ Inverses? _____

Show that order matters in general: for $f(x) = x + 1$ and $h(x) = 2x$, find both compositions.

$f(h(x)) =$ _____ $h(f(x)) =$ _____ Equal? _____

Part 2 — A Pair That Looks Right and Is Not

Test $f(x) = 2x + 1$ and $g(x) = 2x - 1$. Compute, do not assume.

$f(g(x)) =$ _____ Equal to x ? _____ Actual inverse of f : _____

State the lesson in your own words:

Changing a sign is not undoing a function — you must reverse _____.

Part 3 — Non-Linear Pairs and Restrictions

Test $f(x) = x^2$ and $g(x) = \sqrt{x}$. Try $x = 9$ and then $x = -9$.

$f(g(9)) =$ _____ $g(f(9)) =$ _____ $g(f(-9)) =$ _____

So $g(f(x)) = x$ holds only when _____. In general $g(f(x)) = \sqrt{x^2} =$ _____, not x .

Write the complete, correct statement of the inverse relationship for this pair:

f and g are inverses _____

Now test $f(x) = x^3$ and $g(x) = \sqrt[3]{x}$ the same way, at $x = -8$.

$g(f(-8)) =$ _____ Restriction needed? _____ Why the difference from the squaring pair?

Part 4 — Error Analysis

A student is given $f(x) = 2(x - 3)^2 + 4$ and claims $f^{-1}(x) = \sqrt{x - 3} + 4$.

Test the claim at $x = 6$: the claimed inverse gives _____, and f of that is _____.

Undo the steps in reverse: subtract 4, divide by 2, take the root, add 3.

Correct: $f^{-1}(x) =$ _____ Restriction making it a function: _____

Where does the $\pm$ come from, in terms of Day 12? _____

Part 5 — Going Further

1. Verify $f(x) = 5x$ and $g(x) = x/5$ are inverses, both directions. _____

2. Show $f(x) = 1/x$ is its own inverse by composing it with itself.

$f(f(x)) = 1/(1/x) =$ _____ Other self-inverse functions: _____

3. Prove in general that $f(x) = mx + b$ and $f^{-1}(x) = (x - b)/m$ compose to x.

4. If $f(g(x)) = x$ but $g(f(x)) \neq x$, can they still be called inverses? Explain.

No — _____

Exit Ticket

Decide whether $f(x) = 4x - 8$ and $g(x) = (x + 8)/4$ are inverses. Show both compositions, then explain why one direction would not be sufficient proof.

$f(g(x)) =$ _____ $g(f(x)) =$ _____ Inverses? _____